

Basic Derivative Rules

section 2.2
8-29-25

Common notations for derivatives:

$$\frac{dy}{dx}, f'(x), y'$$

Practice: Find Pattern

$$f(x) = x^2 \quad f'(x) = 2x \quad f(x) = x^{10} \quad f'(x) = 10x^9 \quad f(x) = x^{-6} \quad f'(x) = -6x^{-7}$$

↳ 1) exponent as coefficient
2) reduce exponent by 1

POWER RULE

Power rule: $\frac{d}{dx} [x^n] = nx^{n-1}$

Practice:

a) $f(x) = x^3$ $f'(x) = 3x^2$ b) $y = \sqrt{x}$ $y' = \frac{1}{2\sqrt{x}}$
c) $g(t) = \frac{1}{t^4} = t^{-4}$ $g'(t) = -\frac{4}{t^5}$ d) $f(x) = 5$ $f'(x) = 0$

CONSTANT RULE

$$\frac{d}{dx} C = 0$$

PRACTICE

e) $\frac{d}{dx} [5x^3] = 15x^2$

SUM || Difference RULE

$$\frac{d}{dx} [f(x) \pm g(x)] = f'(x) \pm g'(x)$$

Ex:

$$f(x) = 4x^3 - 5x^2 + 7x + 3$$

$$f'(x) = 12x^2 - 10x + 7 + 0$$

Last Rule (very simple)

$$\frac{d}{dx} x = 1$$

PRACTICE

Basically how to avoid quotient rule

Ex1: $y = \frac{5x^5 - 3x^4 + 4}{x}$

$$\frac{5x^5}{x} - \frac{3x^4}{x} + \frac{4}{x} =$$

$$y = 5x^4 - 3x^3 + 4x^{-1}$$

$$y' = 20x^3 - 9x^2 - \frac{4}{x^2}$$

Ex2:

$$y = (x+3)(x^2-1)$$

$$x^3 + 3x^2 - x - 3$$

$$y' = 3x^2 + 6x - 1$$

Ex3: Find the derivative of

$3x^2 + 5x - 1$ and write

tangent line at $(1, 7)$

$$f(x) = 3x^2 + 5x - 1$$

$$f'(x) = 6x + 5$$

$$f'(1) = 11$$

$$m = 11$$

$$\text{point: } (1, 7)$$

$$\text{TAN LINE: } y - 7 = 11(x - 1)$$

Ex4:

find k , $f(x) = x^2 + kx$, so $y = 2x - 9$

1) set func. = to each other.

$$x^2 + kx = 2x - 9$$

2) set derivatives = to each other

$$2x + k = 2$$

3) solve system

$$k = 2 - 2x$$

$$x^2 + (2 - 2x)x = 2x - 9$$

$$x^2 + 2x - 2x^2 = 2x - 9$$

$$-x^2 = -9$$

$$x = \pm 3$$

$$K = 2 - 2x$$

$$K = 2 - 6 = -4$$

$$K = 2 + 6 = 8$$

and thus endeth the notes...